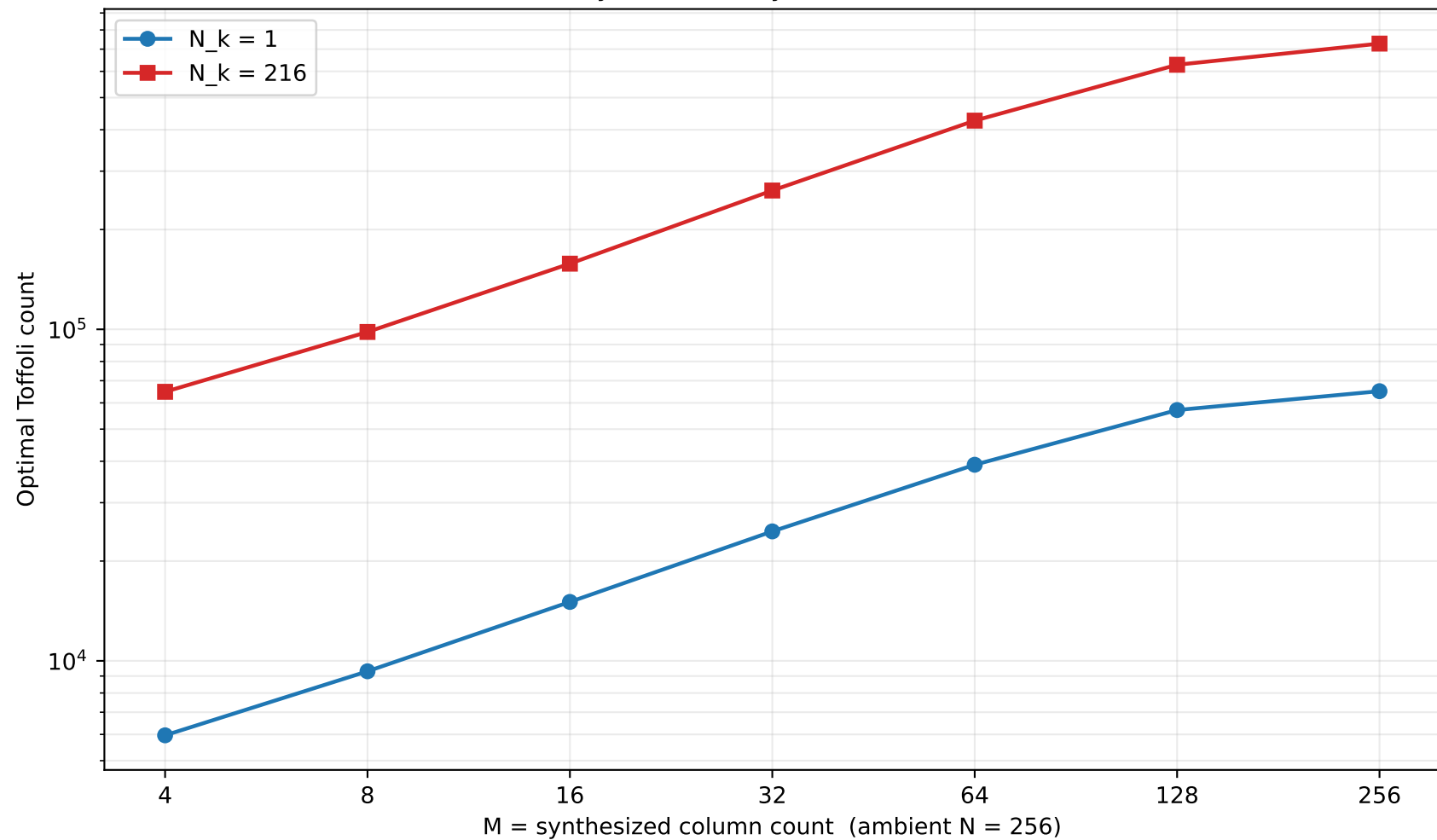


Interferometer isometry: Toffoli vs synthesized columns ($N=256$, $b=32$)



Interferometer isometry optimal Toffoli (ambient $N = 256$, $b = 32$)

M (synth cols)	$N_k=1$ Toffoli	$N_k=216$ Toffoli
1	n/a ($M < 4$)	n/a ($M < 4$)
2	n/a ($M < 4$)	n/a ($M < 4$)
4	5,962	64,801
8	9,301	98,196
16	15,058	157,786
32	24,575	262,254
64	39,040	426,112
128	57,089	628,094
256	65,082	727,560